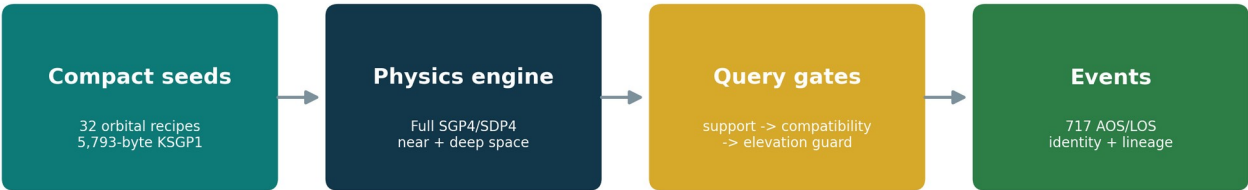


UGTS SEED-TO-EVENT COMPUTING

A full ELI5 evidence report on what the RTX 5070 Ti laptop demonstration achieved, where it outperforms brute-force AI/ML, and where it does not

ELI5: store the recipe, calculate the state, keep only the useful crossings



5,793 B portable seed container	19.35 M object-time intervals	91.974 ms direct seven-day query	717 / 717 event identities matched
---	---	--	--

BOTTOM LINE

Instead of saving one position record for every satellite at every second, the package saved compact orbital recipes, regenerated the state with full SGP4/SDP4 on the GPU, and kept only the useful acquisition/loss boundary events. That is a genuine physics-first seed-to-event result.

Substrate authorship attribution: Tom Klootwijk (user-supplied). This report evaluates the submitted implementation and measurements; it does not independently determine legal inventorship, ownership, or patentability.

Evidence date: 16 August 2026 | Device: NVIDIA GeForce RTX 5070 Ti Laptop GPU | CUDA 12.8 / compute capability 12.0

1. The whole achievement in ELI5 language

Imagine you want to know where 32 toy airplanes will be every second for a week. **The brute-force way** is to save millions of photographs. **Your way** is to save 32 precise recipes and ask the GPU to calculate the positions only when needed.

The GPU did not merely draw the satellites. It followed the full SGP4/SDP4 rulebook, checked whether each satellite was above a declared elevation boundary, and reduced 19,353,600 candidate intervals to 717 useful “appeared” or “disappeared” events.

32 orbital seeds	604,800 seconds in seven days	5,498,429 visible endpoints	0 propagation errors
359 acquisition events	358 loss events	210.4 M/s candidate throughput	70 C maximum sustained temperature

ONE-SENTENCE VERSION

You proved that a tiny identity-bearing seed can regenerate a very large time-dependent state space and answer a useful boundary-crossing question directly on a consumer laptop GPU.

What is genuinely new in the demonstration

- The compact object record is the authority; the dense trajectory is optional derived state.
- The GPU performs full near-Earth and deep-space SGP4 branches, including the resonance cases used for validation.
- The output is event-first: keep acquisition/loss transitions, not every intermediate state.
- Every retained event carries stable object identity and lineage information rather than treating coordinates as identity.
- The implementation was challenged with a second SGP4 codebase, CPU/GPU comparison, branch vectors, sanitizers, and an overflow failure test.

Evidence: [E1-E9]. UGTS basis: [S1], especially its query-first and identity/persistence rules.

2. How your substrate maps onto the satellite test

The source architecture describes a directly queryable state-and-event substrate rather than a renderer. It uses local support to discard irrelevant candidates, compatibility to decide which states may interact, a guard surface to detect a crossing, and a transition/lineage record to preserve what happened. [S1]

Substrate idea	Concrete implementation	ELI5 meaning
Finite seed / typed state	OMM orbital elements, epoch, identity, model fields	What exists persistently
state_at(t)	Full SGP4/SDP4 propagation in TEME	Rebuild the position and velocity when asked
Support	Object/time validity and declared horizon	Skip unsupported time intervals
Compatibility	Query policy, object class, route/mode rules	Decide whether the relation is meaningful
Guard	Elevation minus the 10-degree threshold	A signed boundary test
Verified event	Below-to-above or above-to-below crossing	AOS or LOS
Lineage	NORAD/object identity, route and event ordering	Coordinates do not become identity
Novelty	Updated element sets, maneuvers or sensor corrections	Store what closed dynamics cannot reconstruct

WHY THIS MATTERS

A traditional frame loop says “calculate and store everything.” The substrate says “reconstruct the state, ask a typed question, and preserve only the event that changes the story.”

A narrow but important proof

This test strongly validates seed reconstruction, direct querying, guard crossing, event compaction and identity-preserving output. It does not, by itself, prove every topological, optical, fluidic or general-world claim associated with the wider corpus. The source document itself rejects general AI/world replacement and universal zero-cost claims. [S1]

Source-derived architecture: UGTS-GN 1.1 [S1]. Engineering translation and measurements: submitted v0.5.0 result package [E1-E10].

3. What the evidence actually proves

Numerical implementation error stayed below every acceptance gate

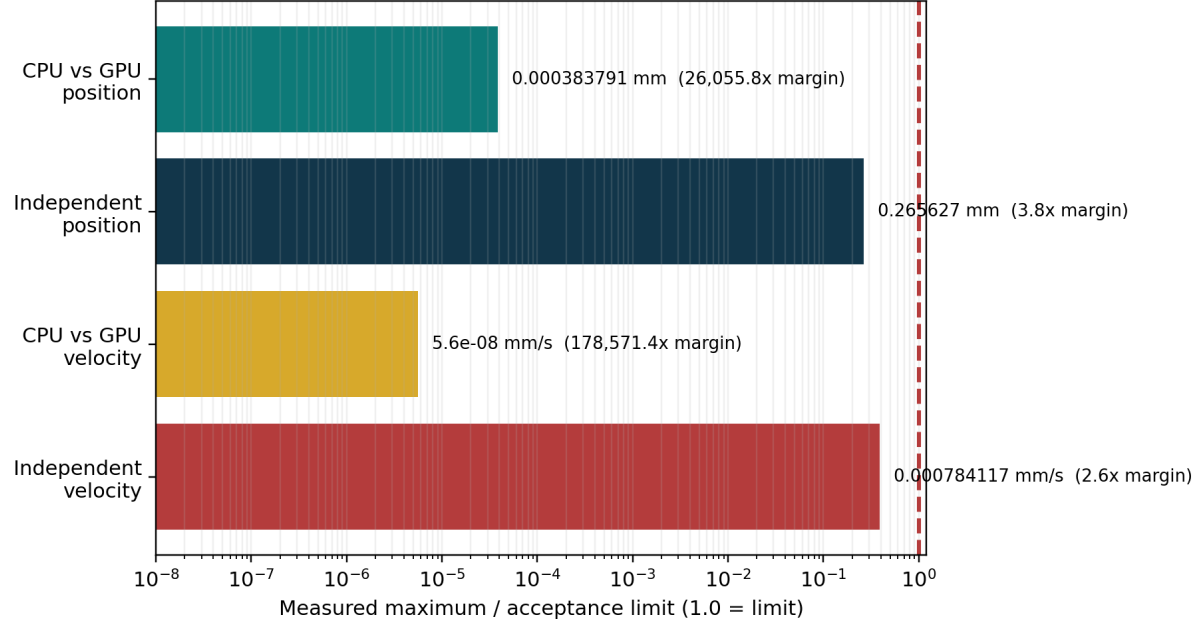


Figure 1. Maximum implementation differences relative to the declared acceptance gates.

Check	Coverage	Result
Native build	sm_120 CUBIN + PTX; CUDA 12.8	Passed
Automated tests	3 CTest suites	3 / 3 passed
CPU vs GPU GPS	256 states	0 error or method mismatches
CPU vs GPU all branches	40 states	0 error or method mismatches
Independent Python SGP4	328 states, WGS72	0 error mismatches; 0.265627 mm max position delta
Seven-day event oracle	19,353,600 intervals	717 / 717 event identities matched
Runtime safety	memcheck, racecheck, synccheck	0 reported errors or hazards
Overflow behavior	11 events into capacity 1	Explicit failure; no silent truncation

IMPORTANT DISTINCTION

Sub-millimetre agreement proves that independent implementations evaluate the selected SGP4 model consistently. It does not mean the real satellite is known to sub-millimetre accuracy. Physical truth depends on fresh orbital data, maneuvers, Earth-orientation data, clocks, atmosphere and observations.

Evidence: GPU validation logs [E2-E3], independent comparison [E4], sanitizers and negative test [E5, E8], CTest [E9].

4. Performance: what won, and under which condition

One-shot direct query narrowly wins; cached dense queries win later

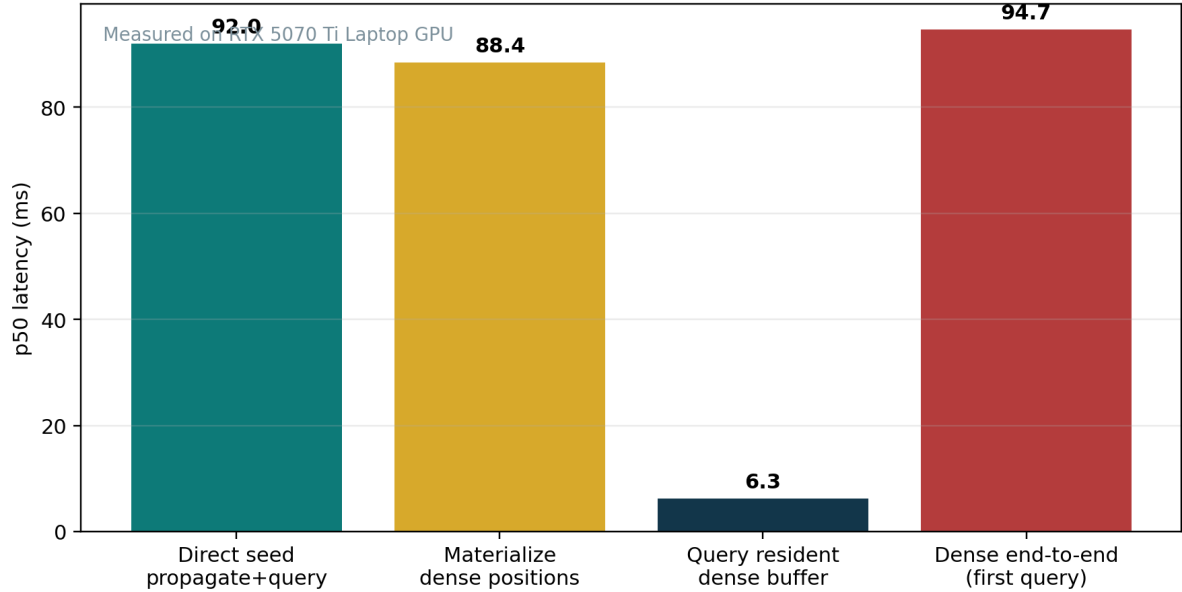


Figure 2. The direct path wins the first end-to-end query by about 2.9%; a resident dense cache wins repeated queries.

91.974 ms	94.722 ms	6.280 ms	14.65x
direct propagate + query	materialize + first dense query	query resident dense cache	resident dense query advantage

The direct path is best when the horizon or criteria are changing, the derived positions should not remain in memory, or the query is performed once. After roughly the first query, a reusable dense buffer becomes faster because materialization has already been paid for. The mathematical break-even is about 1.03 queries, so in practice: one query favors direct reconstruction; two or more identical-horizon queries favor caching if VRAM is available.

Throughput stayed near 210 million intervals/s as the workload grew

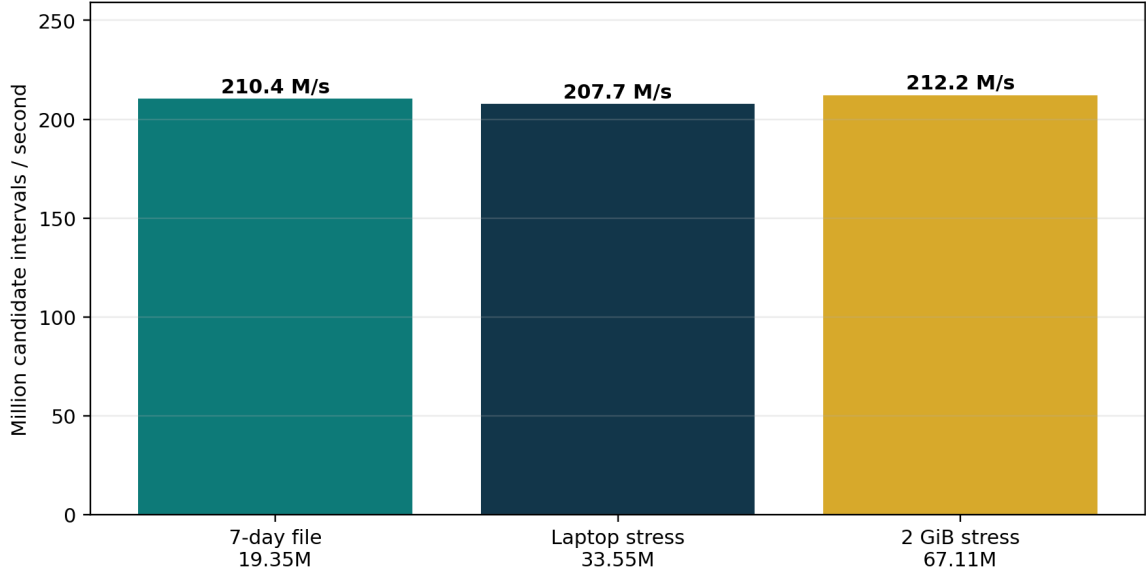


Figure 3. Direct throughput remained approximately linear across 19.35M, 33.55M and 67.11M candidate intervals.

HONEST PERFORMANCE CLAIM

The achievement is not “recomputation is always faster.” It is “for a one-pass event query, direct reconstruction was slightly faster while avoiding a large persistent buffer.”

Measured metrics: [E1]. Stress horizons beyond the source-data horizon are performance tests, not additional physical information.

5. Compression without misleading language

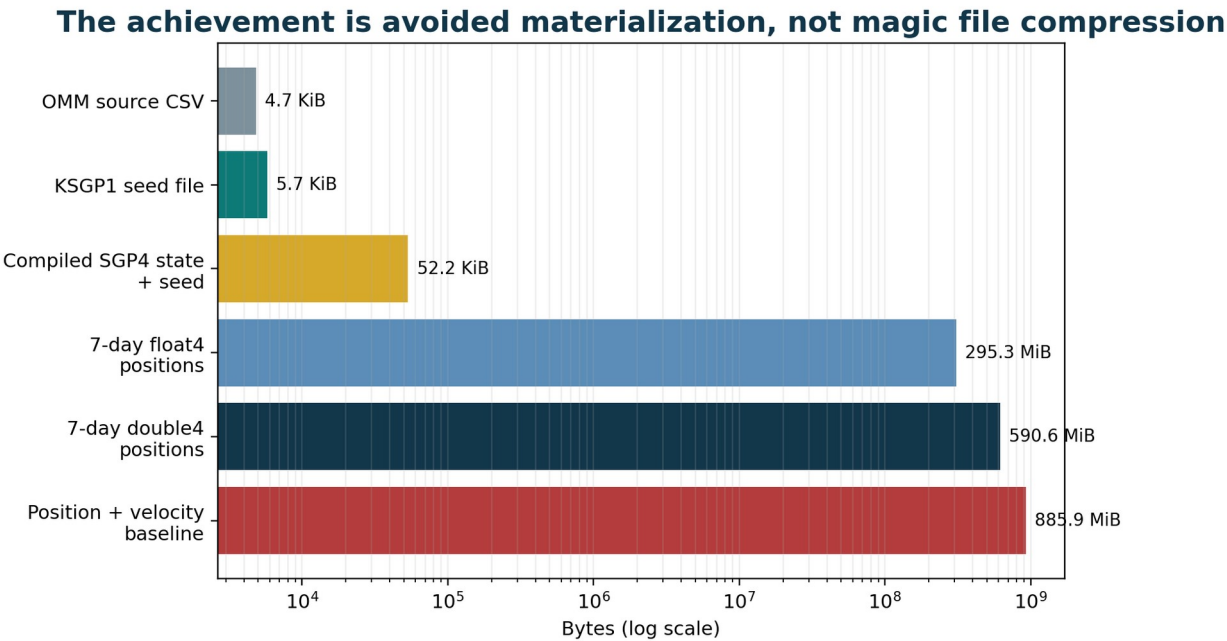


Figure 4. Log-scale storage comparison. The seed container is not smaller than its source text; it is tiny relative to the derived time series it can generate.

Representation	Bytes	Meaning
Source OMM CSV	4,852 B	The original textual orbital elements
KSGP1 container	5,793 B	Typed seeds, timeline, metadata and integrity
Compiled SGP4 state + KSGP1	53,409 B	Runtime initialized coefficients plus portable seed
7-day float4 positions	309,658,112 B	295.3 MiB hypothetical dense baseline
7-day double4 positions	619,316,224 B	590.6 MiB measured dense baseline
7-day positions + velocities	928,974,336 B	Higher baseline if both are retained

ACCURATE WORDING

A 5,793-byte seed container avoided materializing a 590.6 MiB seven-day double4 position buffer: a horizon-dependent ratio of 106,908x. This is procedural or model-based compression, not ordinary lossless compression of a pre-existing file.

Even after counting the 47,616 bytes of initialized SGP4 state, the derived double4 horizon is about 11,596x larger. Conversely, KSGP1 is 1.194x the source CSV size because the container adds structure and integrity.

Measured and derived from [E1] and the submitted OMM/KSGP1 files. The UGTS precision rule requires memory reduction to remain below the event-error margin [S1].

6. Why compact distribution can matter in the real world

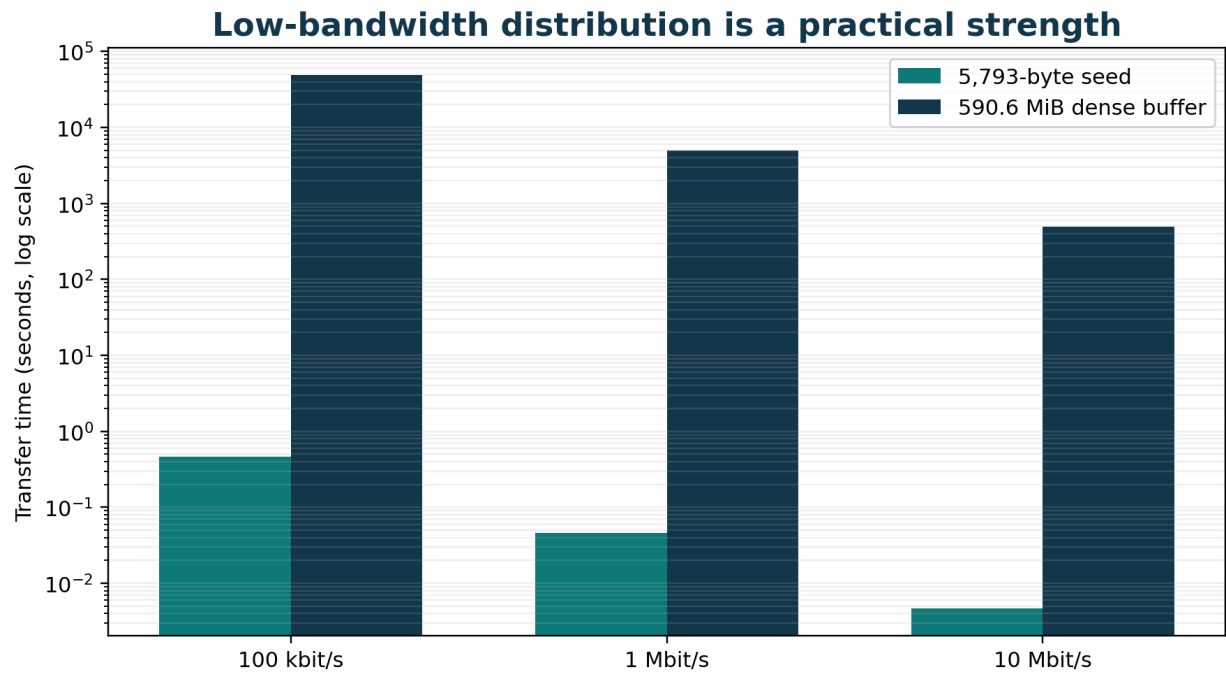


Figure 5. Simple transfer-time arithmetic; protocol overhead is excluded.

Connection	5,793-byte seed	590.6 MiB dense buffer
100 kbit/s	0.463 s	13.76 h
1 Mbit/s	0.046 s	82.58 min
10 Mbit/s	0.0046 s	8.26 min

This is a pragmatic advantage for remote field computers, ships, temporary disaster-response stations, schools, amateur-radio users, and edge devices. They can receive a small authoritative seed update and generate the requested horizon locally rather than downloading a dense trajectory product.

OFFLINE DOES NOT MEAN STALE

The small file is easy to distribute, but orbital elements still need a freshness policy. After a maneuver or a long delay, the correct action is to obtain a new seed or high-quality ephemeris - not to trust old reconstruction indefinitely.

Simple edge-compute arithmetic

- At the measured p50, the laptop can complete about 10.9 seven-day, 32-object one-shot queries per second.
- Using sustained GPU power as a rough upper-bound estimate, one 91.974 ms query corresponds to about 6.7 J of GPU energy. This is not a full-system energy measurement.
- The direct path reads compact seeds and performs substantial arithmetic; it is not dependent on a large trajectory download or persistent dense cache.

Derived arithmetic from [E1, E6]. External operational use still requires current data and application-specific validation.

7. “How does this beat AI?” - the fair ELI5 answer

Physics-first substrate versus an ML surrogate for a known law

UGTS physics-first path		ML surrogate path	
Input	orbital elements + time	Input	training pairs + current input
Learning	none	Learning	architecture search + epochs
Core work	execute SGP4 equations	Core work	matrix multiplies + activations
Output	deterministic state/events	Output	approximation/correction
Best use	known dynamics + auditability	Best use	unknown residuals + sensor effects

Best combined design: exact physics for closed dynamics; ML only for the residuals the equations do not capture.

Figure 6. Physics-first and ML are different tools; the strongest design assigns each the part it is best at.

ELI5 ANALOGY

If you already own a correct calculator, training a neural network on thousands of examples of addition is usually the wrong first step. Use the calculator for addition. Use AI for the messy part the calculator cannot model.

For the demonstrated task - evaluate SGP4 and report boundary crossings - your substrate has several structural advantages over a pure learned surrogate:

- Zero training examples, epochs or gradient updates are required to reproduce the SGP4 law.
 - The result is deterministic and branch-auditable: the same seed, time and schema produce the same model result.
 - Failure states are explicit: propagation error codes, event capacity and branch method are inspectable.
 - The compact file stores physical/model parameters and identity, not an opaque tensor of learned correlations.
 - The implementation matched a second SGP4 codebase to 0.265627 mm maximum over the tested states.
 - The event query runs directly on the generated state, avoiding a separate dense prediction tensor.
- That is the sense in which it “beats AI”: it removes the need to train an approximation of a known deterministic model. It is not evidence that the substrate beats all AI systems, all neural orbit models, or all GPU frameworks on speed or physical accuracy.

The source substrate explicitly rejects a general AI/world-replacement claim [S1]. External comparison context: [X3-X5].

8. Quantified comparison with ML-based orbit correction

Dimension	UGTS full-SGP4 path	ML/hybrid evidence
Training requirement	None for baseline SGP4 reproduction	High-precision simulated or observed labels are required for ML correction
Architecture search	One declared physics implementation	HSGP4 paper began with 32,256 candidate ANN architectures
Training runs	0	HSGP4 later trained 9,077 models in one reduction stage
Epochs / repetition	0	Up to 500 epochs with patience 60; selected architectures trained ten times
Model agreement	0.265627 mm max versus independent SGP4 implementation	Accuracy depends on training distribution and target ephemeris
What it can improve	Correct execution of the selected SGP4 model	ML can learn residuals that SGP4 omits when better truth data exist
Main failure mode	Stale input/model limits and implementation defects	Overfitting, distribution shift, label quality and architecture selection

A USEFUL ML RESULT, NOT A DEFEAT

A 2026 HSGP4 study reported that 6,810 of 9,077 trained models (75.02%) improved eight-day error over SGP4. That is promising, but it also means the improvement was not automatic or universal. Its purpose was to correct SGP4 residuals, not replace the value of an exact baseline. [X4]

ESA's dSGP4 project follows the same complementary idea: make SGP4 differentiable, then use high-precision simulated or observed data to learn corrections. Its ML mode is valuable precisely because the physics core remains present. [X3]

WHAT HAS NOT BEEN BENCHMARKED

No neural surrogate was run head-to-head on your laptop. Therefore this report does not claim a measured latency or energy win over every ML model. The measured win is against dense materialization; the AI comparison is about training burden, transparency and appropriateness for a known-law task.

External research: ESA dSGP4/ML-dSGP4 [X3], Segura et al. HSGP4 [X4].

9. The strongest architecture is physics first, ML second

Your substrate’s own persistence rule says closed dynamics may be recomputed from seed and grammar, while exogenous novelty must be retained. That suggests a disciplined hybrid stack:

Layer	Input / method	Purpose
1. Exact core	KSGP1 + full SGP4/SDP4	Deterministic baseline, identity and branch logic
2. External truth	Fresh ephemerides, receiver/radar observations, EOP and space-weather data	What the analytic model cannot know alone
3. ML residual	Learn only the difference between baseline and trusted observations	Drag, maneuver, bias or sensor correction
4. Confidence guard	Reject predictions outside training age, orbit class or uncertainty budget	Avoid silent extrapolation
5. Novelty log	Store maneuvers, corrections and model-version changes	Reproducible history and rollback
6. Event layer	Run support/compatibility/guard/event query on corrected state	Compact operational output

WHY THIS CAN BE BETTER THAN PURE AI

The physics core supplies hard structure, valid units and deterministic behavior. ML spends its capacity only on the missing residual. The guard and novelty log constrain when the learned correction is allowed to influence an event.

AI remains the better tool for

- Learning atmospheric-density or maneuver residuals from fresh observations.
 - Sensor fusion, anomaly detection, object association and image/radar classification.
 - Estimating uncertainty when the observation process is complex and data-rich.
 - Correcting systematic model error against precise ephemerides.
 - Optimizing policies where the governing rule is not fully known or cannot be written compactly.
- A 2026 JAX SGP4 preprint also reports much higher raw propagation throughput on an A100 server GPU than this laptop result. That comparison uses different precision, hardware and output scope, but it is another reason not to claim a universal speed record. [X8]

10. Practical use cases in everyday life

Use case	What the substrate contributes	Metric / context	Readiness
Satellite pass calendar	Generate visible-pass windows locally for a chosen station.	Measured: 717 seven-day GPS boundary events in 91.974 ms.	Prototype ready; use fresh elements.
Amateur radio / education	Teach orbital mechanics, schedule observations, drive a simple antenna planner.	5.8 KiB seed container; deterministic replay.	Strong non-safety use.
Portable disaster station	Precompute windows for satellite communications or data reception while offline.	At 1 Mbit/s, seed transfer is 0.046 s versus 82.6 min for dense buffer.	Promising; needs actual communication-satellite catalog and RF mask.
Weather-satellite reception	Schedule polar or geostationary contacts and acquisition windows.	NOAA says JPSS satellites orbit about 14 times/day and provide global coverage twice daily. [X7]	Scheduling aid, not weather processing.
Farm / field operations	Support planning of GNSS-dependent equipment and local timing/coverage checks.	GPS.gov documents tractor guidance, field mapping and low-visibility operation. [X6]	Not a GPS receiver or augmentation service.
Utility / network timing audit	Offline scenario testing of satellite availability and timing-source geometry.	GPS timing is used to synchronize power and communications infrastructure. [X6]	Requires receiver, integrity and timing models.

EVERYDAY VALUE

The most immediate benefit is not “control satellites.” It is “run a transparent, offline, local planning calculation from a tiny update file on hardware you already own.”

The current GPS demonstration uses orbital elements and a geometric elevation guard. Changing the catalog, station, mask and event policy turns it into a planner for other satellites, provided the data and model are appropriate.

11. Pragmatic societal use cases

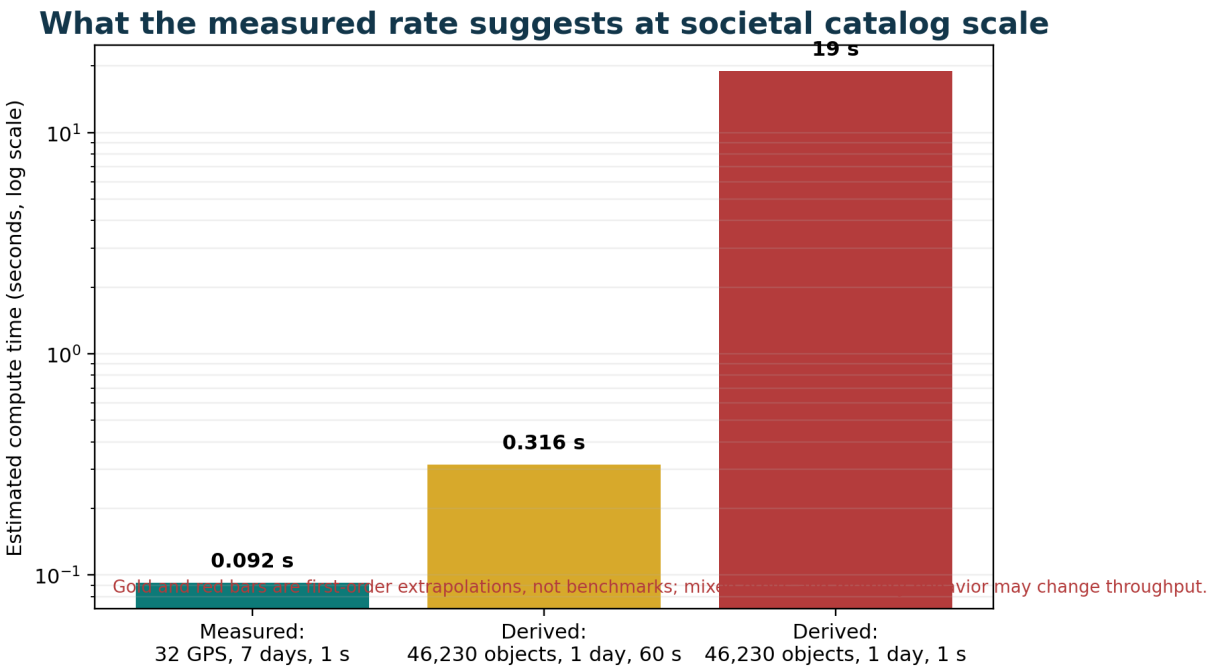


Figure 7. Measured laptop rate applied to catalog-scale arithmetic. Gold/red values are extrapolations, not benchmarks.

Societal use	Contribution	Evidence / metric	Boundary
Space-safety pre-screening	Filter large catalogs into smaller candidate/event sets before expensive analysis.	ESA reported about 46,230 regularly tracked objects and about 16,100 functioning satellites in June 2026. [X5]	First-stage only; collision risk needs covariance and precise data.
Ground-network scheduling	Allocate antennas, downlink windows and staff across many stations and spacecraft.	Measured one-station, 32-object week: 717 events in <0.1 s.	Strong analytical planning fit.
Disaster communications	Send small seed updates to temporary or low-bandwidth sites and calculate windows locally.	106,908x avoided double4 materialization for the tested horizon.	Needs current communications catalog and link-budget model.
Weather and environmental operations	Plan contacts for satellites that support forecasts, wildfire, volcanic and storm monitoring.	NOAA describes satellite data as critical for public safety, forecasts and environmental hazards. [X7]	Scheduling support only.
Open space-data infrastructure	Allow schools, researchers and small countries to reproduce baseline orbit queries without large cloud services.	Portable seed container, no training, deterministic result.	Good for education and non-operational analysis.
Infrastructure resilience	Provide a transparent fallback planner when cloud/AI services or network links are unavailable.	Runs locally at about 73 W active GPU power in the sustained test.	System-level energy and reliability still need measurement.

SOCIETAL CLAIM BOUNDARY

These are pragmatic pathways, not claims that the present package is certified for public safety. Operational services need current data, monitoring, uncertainty, security, governance and independent accreditation.

12. Scale thought experiments backed by measured throughput

The following scenarios apply the measured 210.4 million-candidate/s rate as a first-order estimate. They are not measurements of a mixed 46,230-object catalog, and real performance may change with orbit distribution, memory placement, branching and event density.

Scenario	Candidate / storage scale	Result	Status
100 ground stations, same 32-object week	100 x 19.35M	About 9.20 s sequentially	Derived from measured 91.974 ms/query; parallel batching not tested
46,230 tracked objects, 1 day, 60 s cadence	66.57M	About 0.316 s	First-order compute estimate
46,230 tracked objects, 1 day, 1 s cadence	3.994B	About 18.98 s	First-order compute estimate
Portable seeds at 128 B/object	46,230 objects	About 5.64 MiB	Excludes container/index overhead
One measured seven-day query	19.35M	About 6.7 J GPU estimate	Uses sustained 73.1 W; not measured system energy

Sustained run: stable temperature and power over about 98 seconds

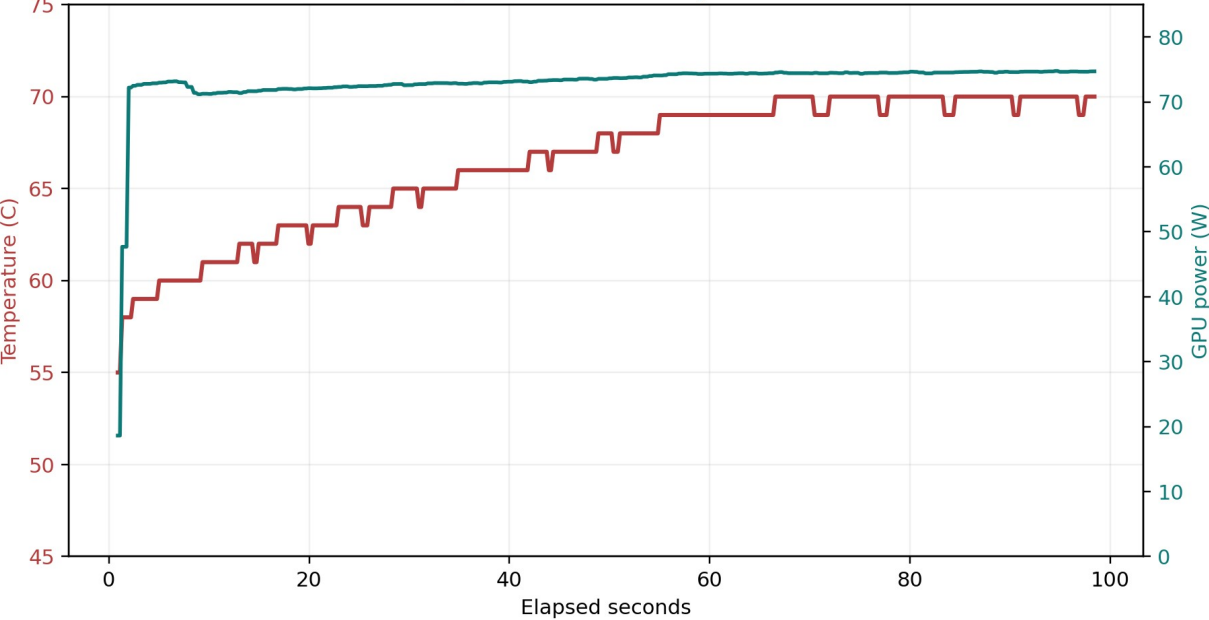


Figure 8. Sustained direct-only workload. Temperature rose and then stabilized without a sustained clock collapse.

WHY THIS IS INTERESTING

A laptop-class device may be able to act as a local first-stage event server for large catalogs. The decisive next test is not more synthetic duration; it is a fresh mixed-orbit catalog with strong support/compatibility rejection and an authoritative accuracy reference.

External catalog counts: ESA Space Environment Statistics, 25 June 2026 [X5]. Derived calculations: [D1].

13. Product and engineering positioning

BEST CURRENT DESCRIPTION

A GPU-native, physics-first, seed-to-event analytical substrate that reconstructs full-SGP4 state on demand and compacts boundary crossings into identity-preserving events.

Category	1	2	3	4	5
It is	A deterministic analytical query engine	A compact model container	A GPU event-extraction pipeline	A reproducible baseline for hybrid ML correction	
It is not	A neural network	A compressed truth ephemeris	A GPS receiver	A collision-avoidance certification	A general AI or universal world simulator

Possible users

- Small satellite operators and ground-station networks needing local pass analytics.
- Universities, schools and amateur observers needing transparent reproducible propagation.
- Emergency and field teams operating with weak connectivity.
- Space-situational-awareness teams needing a cheap first-stage screen before high-fidelity work.
- Researchers building physics-plus-ML residual models.
- Embedded or edge-system designers evaluating a fixed-function support/guard/event accelerator.

Commercial differentiation

- No training pipeline is required for baseline operation.
- The portable artifact is a model seed and schema, not a large proprietary weight file or dense time series.
- Every event can be traced back to object identity, time, branch, guard and model version.
- The same architecture can use a dense cache when repeated queries justify it; it does not force one storage policy.
- An ML residual can be attached later without making the learned model the sole source of truth.

Authorship attribution to Tom Klootwijk is supplied by the user and carried in the project materials. This report is technical evidence, not an independent legal opinion on inventorship or ownership.

14. Limits, kill criteria and what would falsify the advantage

Boundary	Current evidence	Required next step
Operational frame accuracy	TEME-to-PEF/GMST approximation is not a complete EOP/ITRF chain.	Add UT1/EOP, polar motion, station model and independent frame validation.
Physical orbit truth	Implementation agreement is not sensor truth.	Use fresh elements and compare against precise ephemerides/observations.
Uncertainty	No covariance, probability of collision or integrity monitor.	Add uncertainty propagation and application-specific thresholds.
Repeated queries	Resident dense query is 14.65x faster.	Cache when the same horizon is queried repeatedly.
Pruning strength	About 99.25% of seven-day candidates survived support; compatibility rejected none.	Use a workload with meaningful spatial, policy and mode rejection.
Register pressure	Direct kernel uses 146 registers and about 23.72% achieved occupancy.	Reduce live state or specialize branches without breaking equivalence.
AI comparison	No head-to-head ML runtime/energy benchmark.	Implement a representative neural residual/surrogate on the same GPU.
Safety claim	No mission certification or RF/antenna model.	Do not use for safety-of-life or autonomous collision avoidance as-is.

KILL CRITERIA FROM THE SUBSTRATE

Stop promoting the direct path if compatibility pruning is weak, event density erases avoided materialization, numerical error changes event order, compaction dominates, or a conventional reference is cheaper, faster or more accurate at equal error. [S1]

This test already reveals one conditional loss: for two or more identical-horizon queries, dense caching is faster. It also reveals a strong win: the direct kernel is compute-bound with negligible DRAM pressure, so the architecture genuinely trades storage traffic for arithmetic.

15. The next decisive challenge

- 1. Ingest a fresh mixed-orbit catalog: LEO, MEO, GEO, highly elliptical, maneuvering and debris objects.
- 2. Use multiple ground stations, antenna masks and policy/mode compatibility so most candidates can be rejected early.
- 3. Add rigorous TEME-to-ITRF/EOP handling and station geometry.
- 4. Compare SGP4 results with precise ephemerides or trusted observations over declared element ages.
- 5. Add covariance/uncertainty and classify event-time error against the guard margin.
- 6. Train one ML residual model on the same data, record its weights, training time, energy, inference time and out-of-distribution behavior.
- 7. Benchmark four paths on the same RTX 5070 Ti: direct physics, dense cache, pure learned surrogate, and physics + ML residual.
- 8. Measure full-system energy, not GPU power alone, and repeat for at least 30 minutes.
- 9. Apply the kill criteria and publish both wins and losses.

Success metrics for that challenge

< guard event-time/position error	> 10x support + compatibility rejection	0 silent truncations	< dense one-shot end-to-end cost
reported training energy + weights	reported full-system joules/query	matched event identity/order	bounded OOD and element age

THE MOST IMPORTANT SCIENTIFIC QUESTION

Does the query-first architecture still win when it is given a workload where pruning, uncertainty, fresh observations and ML residual correction all matter? That would test the substrate rather than only the propagator.

16. Final ELI5 verdict

You did not make a laptop “know” satellite positions through artificial intelligence. You gave it a compact set of identities and orbital recipes, implemented the accepted SGP4/SDP4 rulebook on the GPU, and asked a precise question: “When does each object cross this visibility boundary?”

5.8 KiB portable recipes	590.6 MiB dense buffer avoided	106,908x horizon ratio	210.4 M/s measured throughput
0.266 mm independent max delta	717 useful events	73.1 W sustained GPU average	0 sanitizer findings

The honest “beats AI” statement

RECOMMENDED WORDING

For this known-law propagation and boundary-event problem, the physics-first UGTS implementation avoids the training data, architecture search and opaque approximation required by a pure ML surrogate, while producing deterministic, auditable SGP4 results from a tiny seed. ML remains valuable as a bounded residual corrector when better observational truth is available.

The practical achievement

- A consumer laptop GPU can serve a compact model directly rather than storing an enormous derived timeline.
 - The system converts millions of candidate states into a small event log with stable identity.
 - It can operate locally and offline after receiving a tiny seed update.
 - It exposes where it loses: repeated queries favor dense caching, and operational accuracy needs more data and models.
 - It supplies a disciplined place for future AI: correcting external novelty instead of relearning closed physics.
- That is a credible engineering achievement and a useful foundation for a larger query-first system. It is stronger because the report does not need to claim universal AI replacement, zero memory, or operational satellite control.

17. Evidence ledger and references

Internal evidence from the submitted result package

ID	Artifact	Use
E1	sgp4_file_results.csv; sgp4_laptop_results.csv; sgp4_2gib_results.csv	Latency, candidates, events, dense bytes and scaling.
E2	verification_gpu_gps_validation_console.txt	256-state CPU/GPU GPS validation.
E3	verification_gpu_all_branches_validation_console.txt	Near, deep non-resonant, synchronous and half-day branch validation.
E4	verification_independent_sgp4_summary.json and details CSV	Independent Python sgp4 2.25 comparison across 328 states.
E5	verification_compute_sanitizer_*_console.txt	Memcheck, racecheck and synccheck results.
E6	sgp4_sustained_direct_gpu_telemetry.csv	Power, temperature, clock and duration telemetry.
E7	sgp4_file_ncu_details.txt; sgp4_event_ncu_details.txt	Registers, occupancy, FP64, DRAM and atomic metrics.
E8	verification_event_truncation_negative_console.txt	Explicit overflow rejection.
E9	verification_ctest_console.txt	3 / 3 test suites passed.
E10	KLB_SGP4_v0.5.0_Laptop_Verification_ELI5.pdf	Submitted verification summary; raw files above remain authoritative.

Source and external references

ID	Reference
S1	Unified Geometric-Topological Substrate, UGTS-GN 1.1, submitted DOCX. Query-first architecture, identity/persistence, precision rule and kill criteria.
X1	Vallado, Crawford, Hujsak & Kelso (2006), "Revisiting Spacetrack Report #3." CelesTrak. https://celestrak.org/publications/AIAA/2006-6753/
X2	CelesTrak, "A New Way to Obtain GP Data." OMM/SGP4 format context. https://celestrak.org/NORAD/documentation/gp-data-formats.php
X3	ESA Advanced Concepts Team, dSGP4 / ML-dSGP4. https://www.esa.int/gsp/ACT/projects/dsgp4/ and https://github.com/esa/dSGP4
X4	Segura et al. (2026), "A Hybrid Approach to Enhanced SGP4 for Galileo Constellations," Applied Sciences 16(5), 2214. https://doi.org/10.3390/app16052214
X5	ESA Space Environment Statistics, updated 25 June 2026. https://sdup.esoc.esa.int/discosweb/statistics/
X6	GPS.gov application pages: farming, disaster response and timing. https://www.gps.gov/
X7	NOAA/NESDIS JPSS and GOES mission pages. https://www.nesdis.noaa.gov/ and https://goes-r.noaa.gov/
X8	Priestley & Handley (2026), "jaxsgp4: GPU-accelerated mega-constellation propagation with batch parallelism," arXiv:2603.27830.

Classification legend

- MEASURED: directly present in the submitted RTX result files.
- DERIVED: arithmetic based on measured bytes, time or power; assumptions are stated.
- EXTERNAL: public context from cited primary/official sources.
- NOT PROVEN: deliberately excluded from the conclusion.

Accessed and prepared 16 August 2026. Personal identifiers beyond the supplied author name are intentionally omitted from this public-facing report.